22.8.26

EXP 4 : SETTING UP A LOCAL AREA NETWORK (LAN)

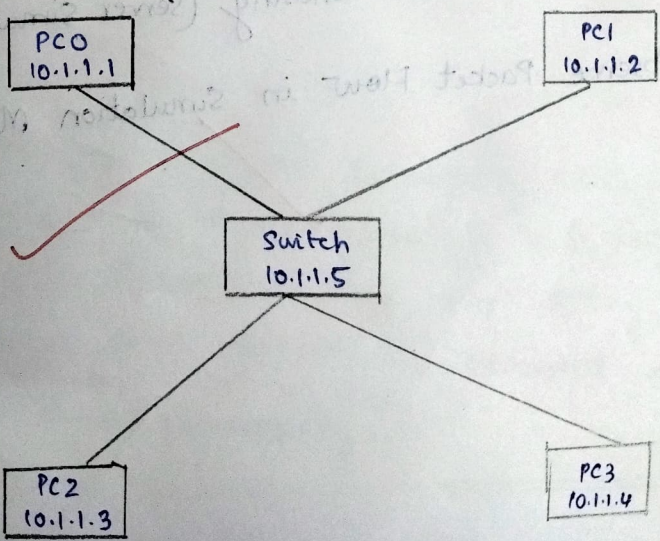
AIM:

To design, set up, and configure a Local Area Network (LAN) connecting four computers through a network switch, assign static IP addresses to all devices, verify connectivity using the ping command, and demonstrate resource/data sharing across the network.

EQUIPMENTS REQUIRED:

S.No.	Equipment	Quantity	Specification
1.	Personal Computers (PC)	4	Ethernet enabled.
2.	Network Switch	1	8/16/24 port, unmanaged/managed
3.	Ethernet cables (Cat5e/Cat6)	4	Straight through, RJ-45 connectors.

NETWORK DIAGRAM:

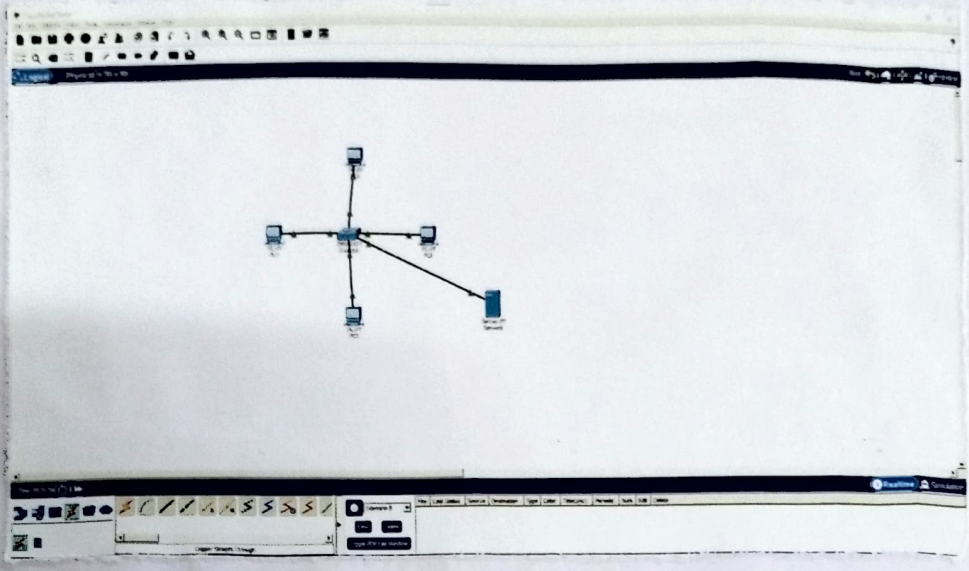


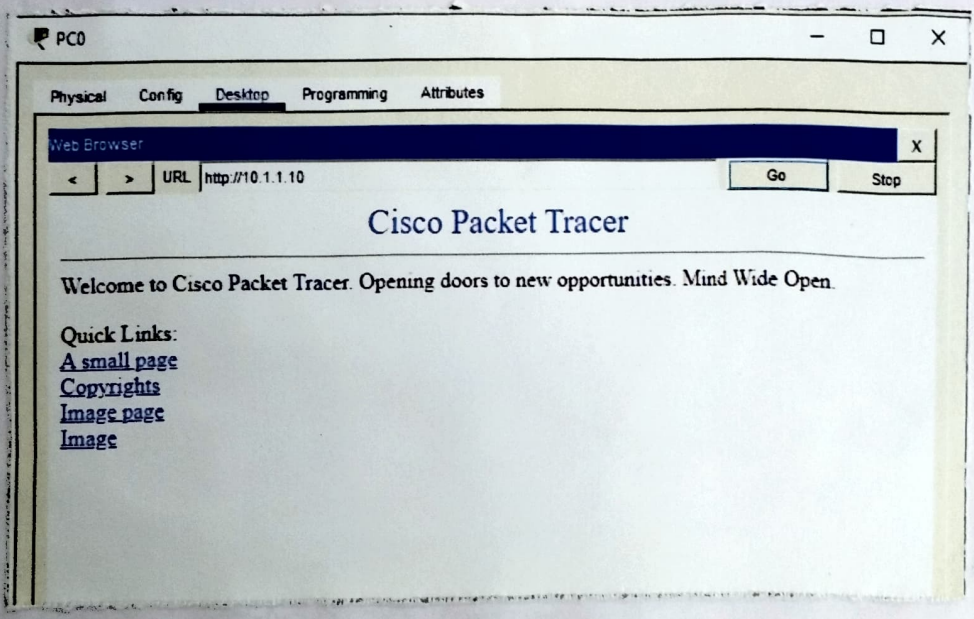
IP ADDRESS CONFIGURATION TABLE:

Device	IP Address	Subnet Mask	Connected To
PC 0	10.1.1.1	255.0.0.0	Switch Port Fa0/1
PC 1	10.1.1.2	255.0.0.0	Switch Port Fa0/2
PC 2	10.1.1.3	255.0.0.0	Switch Port Fa0/3
PC 3	10.1.1.4	255.0.0.0	Switch Port Fa0/4
Switch (Management)	10.1.1.5	255.0.0.0	VLAN 1 Interface

IMPLEMENTATION IN CISCO PACKET TRACER:

1. Place the Devices
2. Cable the Devices
3. Assign IP Addresses to PC's
4. Configure the Switch Management IP (Optional)
5. Verify Connectivity Using Ping
6. Demonstrate Data Sharing (Server Simulation)
7. Observe Packet Flow in Simulation Mode.





RESULT :

All four PC's were successfully connected to the switch in a star topology and assigned unique static IP addresses. Thus, the setting up of a local area network is executed and verified successfully.